APPLIED ECONOMETRICS (PRACTICAL)

STAT 4804

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MULIPLE LINEAR REGRESSION

Multiple linear regression is a generalization of simple linear regression, in the sense that this approach makes it possible to relate one variable with several variables through a linear function in its parameters.

The general mathematical equation for multiple regression is: -

$$y = \hat{a} + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ldots + \beta_n x_n + \text{ə}$$

Following is the description of the parameters used −

- **y** is the response variable.
- $\hat{a}, \beta_1, \beta_2, \beta_3 \ldots \beta_n$ are the coefficients.
- $x_1, x_2, x_3 \ldots x_n$ are the predictor variables.
- ə is the model error (residual), which defines how much variation is introduced in the model while estimating Y.

In multiple linear regression, the relationship between predictors and the outcome variable may not always be a straight line. However, we have the flexibility to shape the relationship by fitting a more appropriate model.

Multiple linear regression computes several key elements to determine the best-fitting line for each predictor. These elements include:

1. Estimated Coefficients: These coefficients quantify the strength and direction of the relationship between each predictor and the outcome variable. They indicate how much the outcome variable is expected to change for a one-unit increase in the corresponding predictor, holding other predictors constant.

2. R-squared (R^2): R-squared represents the proportion of variance in the outcome variable that can be explained by the regression model. It ranges from 0 to 1 and provides a measure of the strength of the relationship between the predictors and the outcome. A higher R-squared indicates a stronger relationship.

3. P-value and T-statistic: These metrics assess the statistical significance of each regression coefficient. The p-value measures the probability of obtaining a coefficient as extreme as observed, assuming that the null hypothesis is true (no relationship between the predictor and the outcome). The t-statistic is calculated as the estimated coefficient divided by its standard error. A higher t-statistic (smaller p-value) suggests a more significant predictor that should be retained in the model, while a low t-statistic (larger p-value) indicates a predictor that may be dropped.

By examining these elements, we can assess the significance and contribution of each predictor in the model and make informed decisions about which predictors to include or exclude based on their statistical significance.

MULTIPLE LINEAR REGRESSION ASSUMPTIONS

To ensure the validity of a multiple linear regression model, it is important to verify the following key assumptions:

1. Normality of Residuals: The residual values (the differences between the observed and predicted values) should follow a normal distribution. This can be assessed by examining a normal probability plot or a histogram of the residuals. Deviations from normality may indicate issues with the model's assumptions.

2. Linearity of Relationships: There should be a linear or curvilinear relationship between the dependent variable and the independent variables. Scatterplots can be used to visualize these relationships. Non-linear patterns may require transformations of variables or the use of non-linear regression techniques.

3. Absence of Multicollinearity: The independent variables should not be highly correlated with each other. High multicollinearity can make it challenging to determine the individual effects of predictors on the dependent variable. To assess multicollinearity, a matrix of Pearson's correlations among the independent variables can be computed. Generally, correlation coefficients should be below 0.8 to avoid severe multicollinearity.

4. Homoscedasticity: Homoscedasticity assumes that the variance of the residual errors is constant across different values of the independent variables. This assumption can be examined by plotting the predicted values against the standardized residual values. If the points are evenly scattered around a horizontal line, homoscedasticity is satisfied.

By checking and satisfying these assumptions, we ensure that the multiple linear regression model provides reliable and valid results. Violations of these assumptions may require further analysis or the use of alternative modeling techniques.

Multiple Linear Regression Analysis in R

In this section, I will be conducting technical implementation of Multiple Linear regression model using *R PROGRAMMING LANGUAGE.*
I will be using *Macroeconomic Data in the Russian Economy* to estimate the *GDP growth rate.*

GDP growth rate refers to the rate at which the Gross Domestic Product (GDP) of a country or region increases over a specific period. GDP is a measure of the total value of all goods and services produced within the boundaries of a country in each period. The GDP growth rate is usually expressed as a percentage and is commonly used to assess the economic health and performance of a country. It indicates the rate of expansion or contraction of the economy. A positive GDP growth rate suggests that the economy is growing, while a negative growth rate indicates a contraction or recession.

Data Set

The dataset contains several macroeconomic time-series regarding the Russian economy. The time-series were collected from the Russian Federal State Statistics Service, the Bank of Russia and Federal Reserve Economic Data. The time-series included in the dataset are:

1. Time: 1-Jan-2005 = 1, every successive step in time represents one quarter
2. Date: Quarterly dates from 1-Jan-2005 to 1-Oct-2021
3. GDP: Quarterly nominal GDP in 2016 prices, excluding seasonal factor (bln RUB)

4. GDPgr: Nominal GDP growth rate (Quarterly, %)
5. M0: Base or high-powered money (bln RUB)
6. M0gr: M0 growth rate (Quarterly, %)
7. BM: M2 measure of money supply (bln RUB)
8. BMgr: M2 growth rate (Quarterly, %)
9. Interest: 90-day interbank rate (APR, %)
10. USDRUB: USD/RUB exchange rate (RUB)
11. EURRUB: EUR/RUB exchange rate (RUB)
12. Unemployment: Unemployment rate (%)
13. PPI: Domestic producer price index (index: 2015=100)
14. PPIgr: Growth rate of producer price index (Quarterly, %)
15. OIL : Spot prices of Brent per barrel (USD)
16. OILgr: Growth rate of Brent prices (Quarterly, %)
17. WAGE: Average monthly nominal wage rate (RUB)
18. WAGEgr: Changes in nominal wage rate (Quarterly, %)
19. CPI: Change in CPI as a ratio (End of quarter to end of previous quarter, %)
20. Inflation: Percentage change in CPI, calculated as Relative CPI - 100 (Quarterly, %)

The data was used to in time-series regression modelling to explain the factors affecting inflation in Russia. Some other modelling ideas for the dataset are:

1. Shift the focus from factor analysis to predicting future inflation.
2. Perform factor analyses of other key macroeconomic variables, such as the GDP growth rate, the unemployment rate, or the interest rate.

Due to the low number of available observations because of quarterly sampling, this dataset is probably better suited to time-series econometric analysis rather than more modern machine learning methods.

Data Exploration

Let's have a quick overview of the dataset so we can apply relevant processing before fitting the model.

```
#Loading the Data Set in R Console

data_set <- read.csv(

  "C:\\Users\\Qura tul Ain\\Documents\\FORM\\Xyed's Semester
8th\\Projects\\Econ_porject\\data.csv"

)


# Overview the first 6 observations

head(data_set)


#Checking the dimension of a data set

dim(data_set)
```

```
> head(data_set)
  ï..Time       Date      GDP    GDPgr       M0      M0gr       BM  BMgr Interest
1       1 01.01.2005 15883.74 1.718049 1534.755 18.636706 5288.740 14.23     4.56
2       2 01.04.2005 16125.54 1.499469 1481.698 -3.457034 5487.113  3.75     6.39
3       3 01.07.2005 16409.70 1.731689 1650.743 11.408870 6006.595  9.47     5.66
4       4 01.10.2005 16748.21 2.021147 1740.685  5.448577 6442.792  7.26     7.10
5       5 01.01.2006 17071.55 1.894039 2009.240 15.428122 7213.379 11.96     5.80
6       6 01.04.2006 17400.20 1.888744 1928.809 -4.003056 7374.657  2.24     5.48
   USDRUB  EURRUB Unemployment      PPI    PPIgr   OIL     OILgr  WAGE     WAGEgr
1 27.7487 37.8409     8.200000 40.33314 2.212723 53.22 31.797920  7638  0.7385914
2 27.8548 36.0274     7.400000 41.48579 2.857834 55.36  4.021045  8234  7.8030898
3 28.6282 34.6258     7.166667 42.12481 1.540326 61.70 11.452312  8674  5.3436969
4 28.5348 34.3074     7.466667 42.71607 1.403597 58.34 -5.445705  9651 11.2635462
5 28.4821 34.3352     7.766667 43.86848 2.697827 66.06 13.232773  9397 -2.6318516
6 27.6996 33.6273     7.433333 45.46317 3.635173 73.20 10.808356 10401 10.6842609
     CPI Inflation
1 105.27      5.27
2 102.58      2.58
3 100.57      0.57
4 102.12      2.12
5 104.98      4.98
6 101.11      1.11
> dim(data_set)
[1] 68 20
```

From the previous results, we can observe that the data set has 68 observations and 20 columns.

However, based on the problem statement, we do not need the *Time and Date* column because we are now dealing with a regression problem.

```
#Deleting the time and date column

data_set <- subset(data_set, select = -c(ï..Time, Date))

head(data_set)

dim(data_set)
```

```
> #Deleting the time and date column
> data_set <- subset(data_set, select = -c(ï..Time, Date))
> head(data_set)
       GDP    GDPgr       M0      M0gr       BM  BMgr Interest  USDRUB  EURRUB
1 15883.74 1.718049 1534.755 18.636706 5288.740 14.23     4.56 27.7487 37.8409
2 16125.54 1.499469 1481.698 -3.457034 5487.113  3.75     6.39 27.8548 36.0274
3 16409.70 1.731689 1650.743 11.408870 6006.595  9.47     5.66 28.6282 34.6258
4 16748.21 2.021147 1740.685  5.448577 6442.792  7.26     7.10 28.5348 34.3074
5 17071.55 1.894039 2009.240 15.428122 7213.379 11.96     5.80 28.4821 34.3352
6 17400.20 1.888744 1928.809 -4.003056 7374.657  2.24     5.48 27.6996 33.6273
  Unemployment      PPI    PPIgr   OIL     OILgr  WAGE     WAGEgr    CPI Inflation
1     8.200000 40.33314 2.212723 53.22 31.797920  7638  0.7385914 105.27      5.27
2     7.400000 41.48579 2.857834 55.36  4.021045  8234  7.8030898 102.58      2.58
3     7.166667 42.12481 1.540326 61.70 11.452312  8674  5.3436969 100.57      0.57
4     7.466667 42.71607 1.403597 58.34 -5.445705  9651 11.2635462 102.12      2.12
5     7.766667 43.86848 2.697827 66.06 13.232773  9397 -2.6318516 104.98      4.98
6     7.433333 45.46317 3.635173 73.20 10.808356 10401 10.6842609 101.11      1.11
> dim(data_set)
[1] 68 18
```

We have data set reduced to 18 columns.

Testing of Multiple Linear Regression Model

Equation: -

$$y = \hat{a} + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ldots + \beta_n x_n + \varepsilon$$

```
#Fitting the Multiple Linear regression Model

gdp_model <- lm(formula = GDPgr ~ GDP + M0 + M0gr + BM + BMgr
+ Interest +USDRUB + EURRUB + Unemployment + PPI + PPIgr + OIL
+ OILgr + WAGE + WAGEgr + CPI +Inflation, data = data_set)

options(scipen = 0)

summary(gdp_model)
```

```
Call:
lm(formula = GDPgr ~ GDP + M0 + M0gr + BM + BMgr + Interest +
    USDRUB + EURRUB + Unemployment + PPI + PPIgr + OIL + OILgr +
    WAGE + WAGEgr + CPI + Inflation, data = data_set)

Residuals:
     Min       1Q   Median       3Q      Max
-2.95187 -0.37750 -0.02337  0.52305  1.59525

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.713e+01  1.452e+01  -1.180 0.243497
GDP           1.668e-04  3.315e-04   0.503 0.617018
M0            3.823e-04  5.058e-04   0.756 0.453219
M0gr          6.478e-02  4.040e-02   1.603 0.115040
BM            6.808e-05  1.843e-04   0.369 0.713302
BMgr         -7.547e-02  7.094e-02  -1.064 0.292446
Interest     -2.883e-01  7.011e-02  -4.113 0.000143 ***
USDRUB        1.406e-01  8.702e-02   1.616 0.112281
EURRUB        1.516e-02  7.017e-02   0.216 0.829840
Unemployment  2.706e-01  3.250e-01   0.833 0.408885
PPI          -5.637e-02  3.773e-02  -1.494 0.141261
PPIgr         2.056e-02  6.509e-02   0.316 0.753342
OIL           4.030e-02  1.384e-02   2.912 0.005313 **
OILgr        -4.920e-02  6.500e-03  -7.569 6.88e-10 ***
WAGE         -2.451e-04  1.644e-04  -1.491 0.142118
WAGEgr        6.494e-02  4.356e-02   1.491 0.142174
CPI           1.234e-01  1.331e-01   0.927 0.358383
Inflation            NA         NA      NA       NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 51 degrees of freedom
Multiple R-squared:  0.7237,    Adjusted R-squared:  0.637
F-statistic: 8.349 on 16 and 51 DF,  p-value: 2.491e-09
```

Interpretations of coefficients β

The table Coefficients gives the estimate for each parameter (column Estimate), together with the p-value of the nullity of the parameter (column Pr(>|t|)).

The Test Hypothesis are:-

- H_0 ; $\beta_j = 0$
- H_1 ; β_j != 0

In practice, usually conditions/assumptions are checked before interpreting the coefficients of model. However, I present the interpretations before testing the conditions to get clear in interpreting the results, and less about finding a valid model.

Conclusion:-

Based on the output of our model, we conclude that:

- Significant and positive relationships exist between GDPgr and the following variables: GDP, M0, M0gr, BM, Unemployment, USDRUB, EURRUB, PPIgr, OIL, WAGEgr and CPI. For a one-unit increase in GDP, GDPgr increases by an average of 1.668e-04 (p-value = 0.617018). Similarly, for a one-unit increase in M0, M0gr, BM, Unemployment, USDRUB, EURRUB, PPIgr, OIL, WAGEgr and CPI, GDPgr increases by 3.823e-04, 6.478e-02, 6.808e-05, 2.706e-01, 1.406e-01, 1.516e-02, 2.056e-02, 4.03e-02, 6.494e-02 and 1.234e-01 respectively. (p-values = 0.453219, 0.11504, 0.713302, 0.408885, 0.112281, 0.829840, 0.753342, .0053130, 0.142174, 0.358383).
- Significant and negative relationships exist between GDPgr and the following variables:
  BMgr, Interest, PPI, OILgr, WAGE. For one unit increase in BMgr, GDPgr decreases by an average of 7.547e-02 (p-value = 0.292446). Similarly, for one unit increase in Interest, PPI, OILgr, and WAGE, GDPgr decreases by 2.883e-01, 5.637e-02, 4.920e-02, 2.451e-04 respectively (p-value = 0.000143, 0.141261, 6.88e-10, 0.142118).
- Intercept value shows negative significant relationship of -1.713.
- However, it is interesting to note that Inflation variable does not show any significant relationship to GDPgr.
- The p-value = 2.491e-09. The null hypothesis is rejected, so we conclude that our model is better than a model with only the intercept because at least one coefficient β is significantly different from 0.
- $\boldsymbol{R^2}$ • for this model is 0.7237, which means that 72.37% of the GDPgr of the Russian Economy is explained by all the defined (except for Inflation variable)independent variables in the given data set.

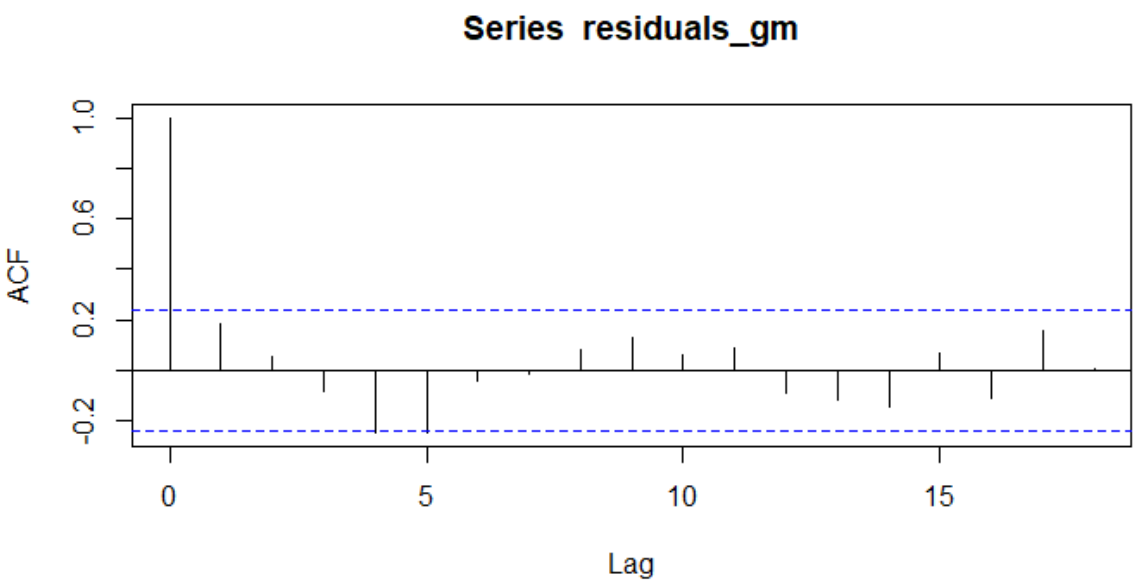
As for simple linear regression, multiple linear regression requires some conditions of application for the model to be usable and the results to be interpretable.

Some of the Important Conditions/Assumptions include:

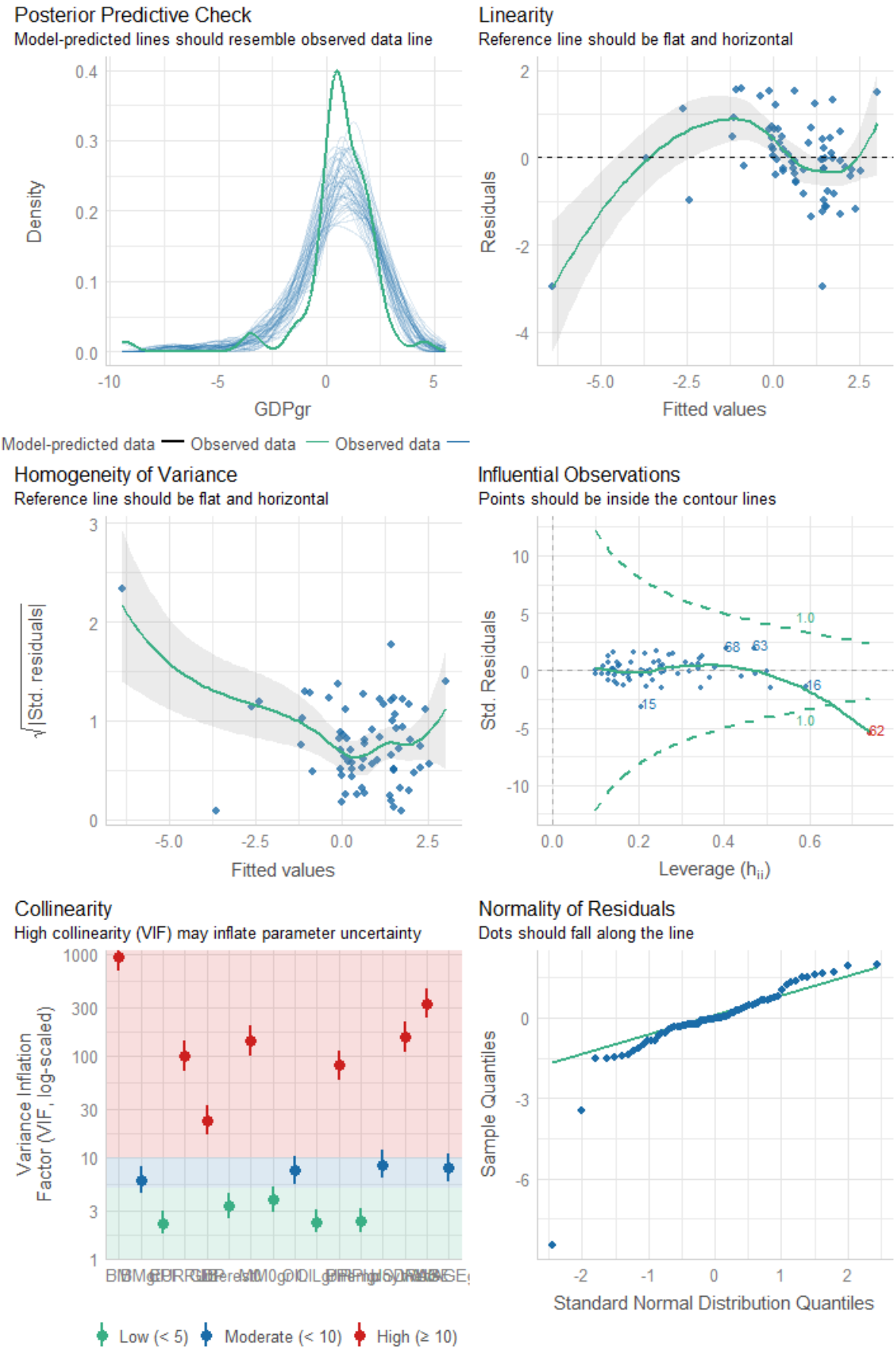
- **Linearity** of the relationships between the dependent and independent variables.
- **Independence** of observations.
- **Normality** of the residuals.
- **Homoscedasticity** of the residuals.
- No influential points **Outliers**.
- No **Multicollinearity**: Multicollinearity arises when there is a strong linear correlation between the independent variables, conditional on the other variables in the model.

Checking the Assumptions/Conditions for the above Model in R programming.

```
#Examining the Assumptions of Multiple Linear Regression Model
residuals_gm <- gdp_model$residuals
library(lmtest)
library(performance)
check_model(gdp_model)
acf(residuals_gm, type = "correlation")
dwtest(gdp_model)
```




```
Durbin-Watson test

data:  gdp_model
DW = 1.5929, p-value = 5.498e-10
alternative hypothesis: true autocorrelation is greater than 0
```


Posterior Predictive Check
Model-predicted lines should resemble observed data line
Density
GDPgr
Model-predicted data
Observed data
Observed data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Low (< 5)
Moderate (< 10)
High (≥ 10)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
Standard Normal Distribution Quantiles

Based on the diagnostic plots above, we see that:

- Linearity assumption is seriously violated. (Top right plot)
- Homogeneity of variances is also not respected. (Middle left plot)
- Only one outlier exists in the model. (Middle right plot)
- High Multicollinearity among the independent variables exist. (Bottom left plot)
- Normality of residuals seem to be respected. (Bottom right plot)
- The Durbin-Watson test yielded a very small p-value of 5.498e-10, indicating strong evidence to reject the null hypothesis of no autocorrelation in the residuals. Therefore, we conclude that there is significant autocorrelation present. The alternative hypothesis suggests a positive correlation between the residuals of adjacent observations. In summary, the Durbin-Watson test confirms the presence of autocorrelation in the residuals, indicating a lack of independence between adjacent observations.

There are some serious violations of Assumptions for the Model 1, even though the $\boldsymbol{R^2}$ value for the model is high (0.7237), the Model 1 is not a good fit for the GDPgr.

Developing another model (Model 2), using Step Wise Regression Model technique to develop variables solely based on calculations, in R.

Multiple Linear Regression Model 2

```
#Applying step wise regression model to finding best variables
to fit the model.

gdp_model_2 <- step(lm(GDPgr ~ ., data = data_set), direction
= "both")

summary(gdp_model_2)
```

```
> summary(gdp_model_2)

Call:
lm(formula = GDPgr ~ M0 + M0gr + BMgr + Interest + USDRUB + PPI +
    OIL + OILgr + WAGE + WAGEgr + CPI, data = data_set)

Residuals:
     Min       1Q   Median       3Q      Max
-3.05029 -0.41219 -0.03426  0.57968  1.75383

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.596e+01  1.201e+01  -1.330 0.189034
M0           5.829e-04  3.205e-04   1.819 0.074279 .
M0gr         6.167e-02  3.830e-02   1.610 0.112978
BMgr        -8.941e-02  6.598e-02  -1.355 0.180790
Interest    -2.932e-01  5.001e-02  -5.863 2.53e-07 ***
USDRUB       1.769e-01  3.549e-02   4.983 6.35e-06 ***
PPI         -6.267e-02  3.443e-02  -1.820 0.074061 .
OIL          4.195e-02  1.053e-02   3.982 0.000199 ***
OILgr       -4.925e-02  5.661e-03  -8.700 5.52e-12 ***
WAGE        -1.924e-04  8.286e-05  -2.322 0.023883 *
WAGEgr       4.707e-02  2.271e-02   2.072 0.042861 *
CPI          1.565e-01  1.184e-01   1.323 0.191343
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.02 on 56 degrees of freedom
Multiple R-squared:  0.7169,    Adjusted R-squared:  0.6613
F-statistic: 12.89 on 11 and 56 DF,  p-value: 1.012e-11
```

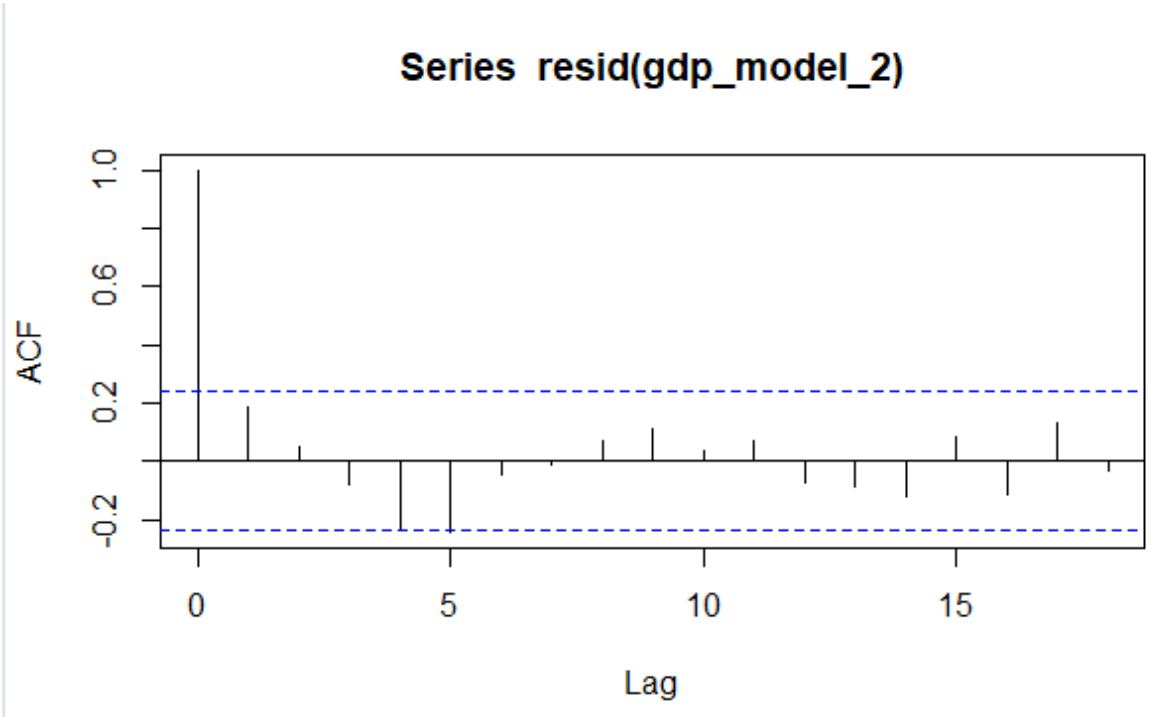
Checking the Assumptions/Conditions for the above Model in R programming.

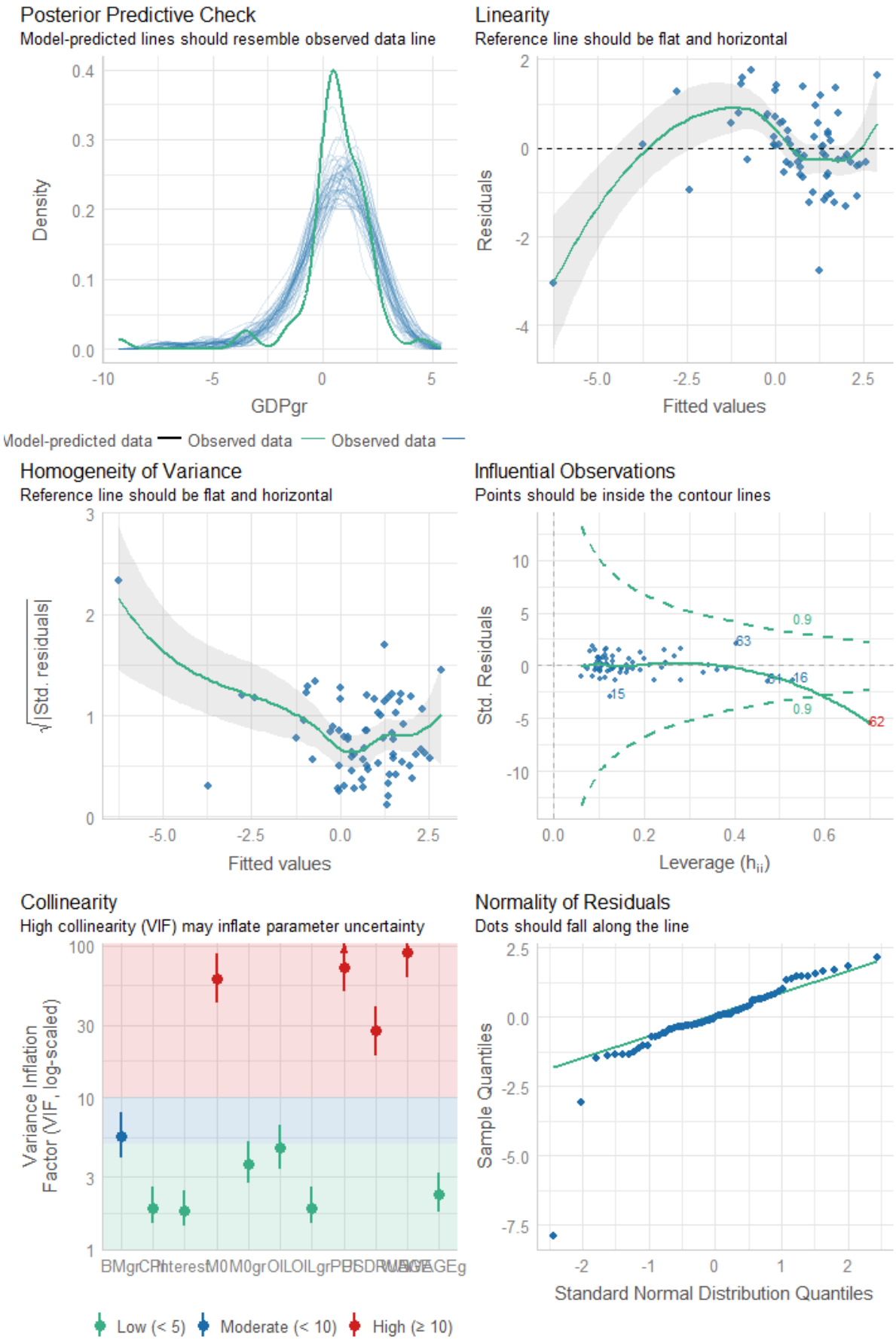
```
#Examining the Assumptions of 2nd Multiple Linear Regression Model

library(performance)

library(lmtest)


acf(resid(gdp_model_2), type = "correlation")

dwtest(gdp_model_2)

check_model(gdp_model_2)
```

```
Durbin-Watson test

data:  gdp_model_2
DW = 1.5887, p-value = 0.01344
alternative hypothesis: true autocorrelation is greater than 0
```

Based on the diagnostic plots below, we see that:

- Linearity assumption is not met while removing some variables. (Top right plot)
- Homogeneity of variances is still not respected. (Middle left plot)
- Only one outlier exists in the model. (Middle right plot)
- High Multicollinearity among the M0, PPI, USDRUB, and WAGE. (Bottom left plot)
- Normality of residuals seem to be respected. (Bottom right plot)
- The p-value associated with the Durbin-Watson test is 0.01344, which is below the conventional significance level of 0.05. Therefore, we do have sufficient evidence to reject the null hypothesis. The null hypothesis states that there is no autocorrelation in the residuals. The alternative hypothesis suggests that there is autocorrelation, specifically a positive correlation between the residuals of adjacent observations. However, since the p-value is less than 0.05, we have enough evidence to support the presence of significant positive autocorrelation.




Posterior Predictive Check
Model-predicted lines should resemble observed data line
Density
GDPgr
Model-predicted data
Observed data
Observed data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
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Points should be inside the contour lines
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Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Low (< 5)
Moderate (< 10)
High (≥ 10)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
Standard Normal Distribution Quantiles

Interpretations of coefficients β for Model 2

The table Coefficients gives the estimate for each parameter (column Estimate), together with the p-value of the nullity of the parameter (column Pr(>|t|)).

The Test Hypothesis are:-

- H_0 ; β_j= 0
- H_1 ; β_j != 0

Conclusion:-

Step wise Regression Model removed some of the variables solely based on calculation in R studio. Based on the output of our model, we conclude that:

- Significant and positive relationships exist between GDPgr and the following variables: M0, M0gr, USDRUB, OIL, WAGEgr, and CPI. For a one-unit increase in M0, GDPgr increases by an average of 5.93e-04 (p-value = 0.074). Similarly, for a one-unit increase in M0gr, USDRUB, OIL, WAGEgr and CPI, GDPgr increases by 6.17e-02, 1.769e-01, 4.195e-02, 4.71e-2, and 1.565e-01 respectively. (p-values = 0.112978, 0.180790, 6.35e-06, 0.000199, 0.042861, 0.191343).
- Significant and negative relationships exist between GDPgr and the following variables:
  BMgr, Interest, PPI, OILgr, and WAGE. For one unit increase in BMgr, GDPgr decreases by an average of 8.94e-02 (p-value = 0.18079). Similarly, for one unit increase in Interest, PPI, OILgr, and WAGE, GDPgr decreases by 2.53e-07, 6.267e-02, 4.925e-02, and 1.924e-4 respectively (p-value = 2.53e-07, 0.07, 5.52e-12, 0.023).
- Intercept value shows negative significant relationship of 1.596e+01. (p-value = 0.189)
- The p-value = 1.012e-12. The null hypothesis is rejected, so we conclude that our model is better than a model with only the intercept because at least one coefficient β is significantly different from 0.
- $\boldsymbol{R^2}$ for this model is 0.7169, which means that 71.69% of the GDPgr of the Russian Economy is explained by independent variables applied in the Model 2 from the given data set.

Model 2 seriously violates linearity, Homogeneity, Collinearity assumptions etc. Therefore, applying another model now based on theoretically dependence of GDPgr on the variables given in the data set.

Multiple Linear Regression Model 3

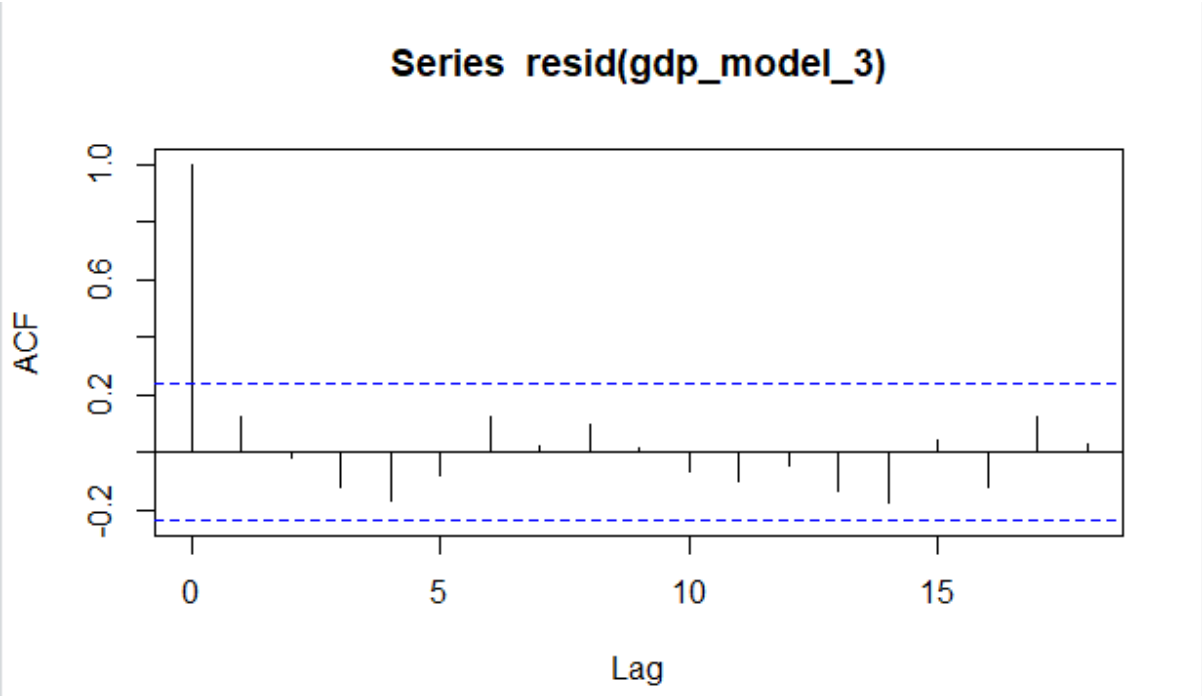
```
#Reducing data to theoretical related variables.

reduced_data_set <- subset(data_set, select = c(GDPgr, BMgr,
CPI, Interest, M0gr, OILgr, WAGEgr, Unemployment, PPIgr,
EURRUB))

#Applying 3rd Model

gdp_model_3 <- lm(GDPgr ~ . , data = reduced_data_set)

summary(gdp_model_3)
```

```
> summary(gdp_model_3)

Call:
lm(formula = GDPgr ~ ., data = reduced_data_set)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.4716 -0.4543  0.1605  0.5075  2.1296 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   0.725385  13.581687   0.053   0.9576    
BMgr          0.036251   0.069810   0.519   0.6055    
CPI          -0.004075   0.137761  -0.030   0.9765    
Interest     -0.259721   0.060488  -4.294 6.77e-05 ***
M0gr          0.029961   0.042896   0.698   0.4877    
OILgr        -0.035641   0.005771  -6.176 6.97e-08 ***
WAGEgr        0.027312   0.020459   1.335   0.1871    
Unemployment  0.235143   0.171133   1.374   0.1747    
PPIgr         0.144269   0.055788   2.586   0.0122 *  
EURRUB        0.009781   0.010921   0.896   0.3741    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.189 on 58 degrees of freedom
Multiple R-squared:  0.6014,	Adjusted R-squared:  0.5396 
F-statistic: 9.724 on 9 and 58 DF,  p-value: 6.941e-09
```

Checking the Assumptions/Conditions for the above Model in R programming.

```
#Examining the Assumptions of 3rd Multiple Linear Regression
Model

library(performance)

library(lmtest)


acf(resid(gdp_model_3), type = "correlation")

dwtest(gdp_model_3)

check_model(gdp_model_3)
```

```
Durbin-Watson test

data:  gdp_model_3
DW = 1.7506, p-value = 0.09631
alternative hypothesis: true autocorrelation is greater than 0
```



Based on the diagnostic plots below, we see that:

- Linearity assumption is somewhat linear with this model. (Top right plot)
- Homogeneity of variances is still not respected. (Middle left plot)
- Only one outlier exists in the model. (Middle right plot)
- VIF is less than or equal to 5 indicate that Collinearity is not an issue in this model. (Bottom left plot)

- Normality of residuals is respected. (Bottom right plot)
- The p-value associated with the Durbin-Watson test is 0.0963, which is above the conventional significance level of 0.05. Therefore, we do not have sufficient evidence to reject the null hypothesis. The null hypothesis states that there is no autocorrelation in the residuals. The alternative hypothesis suggests that there is autocorrelation, specifically a positive correlation between the residuals of adjacent observations. However, since the p-value is greater than 0.05, we do not have enough evidence to support the presence of significant autocorrelation.

Interpretations of coefficients β for Model 3

The table Coefficients gives the estimate for each parameter (column Estimate), together with the p-value of the nullity of the parameter (column Pr(>|t|)).

The Test Hypothesis are:-

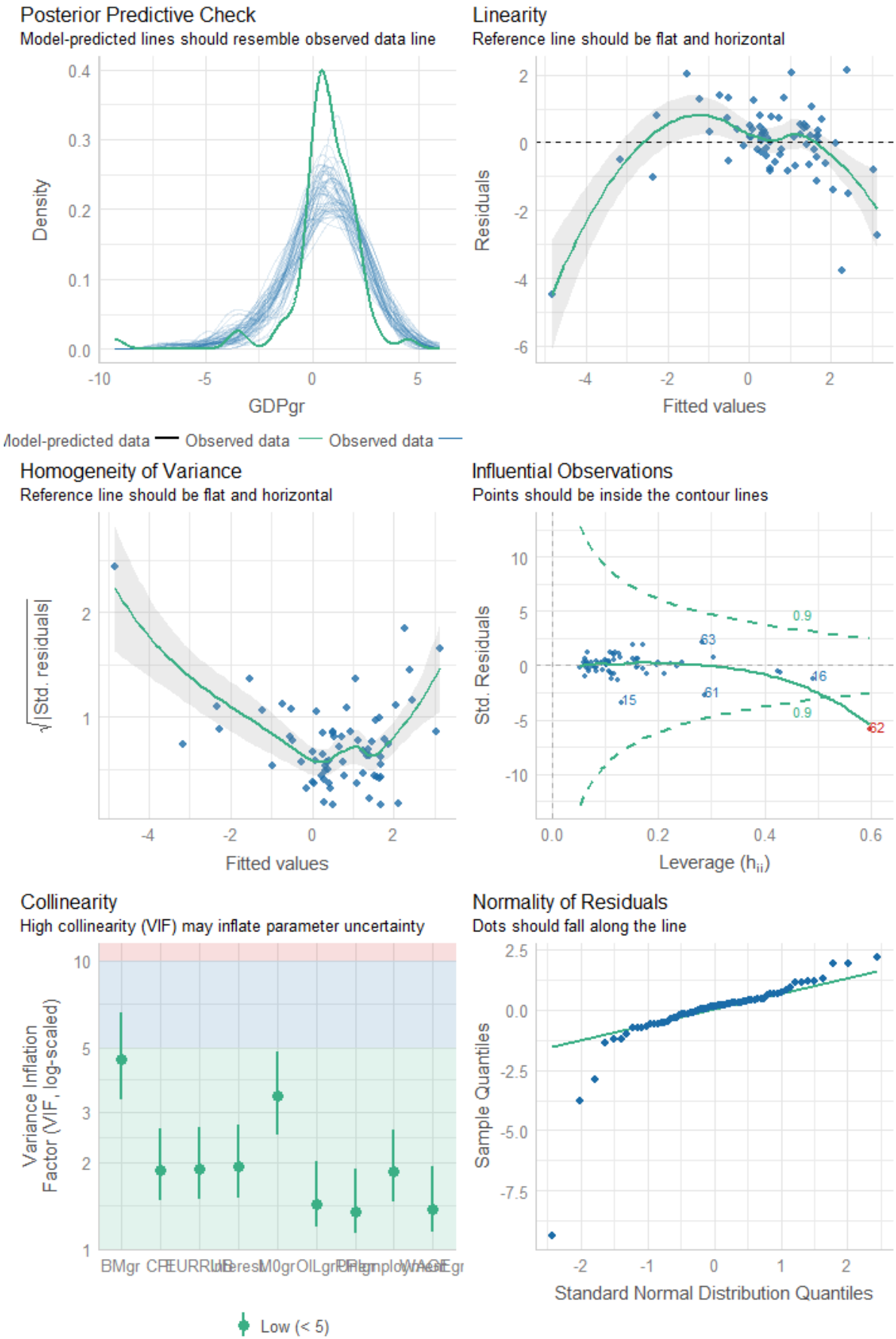
- H_0 ; $\beta_j = 0$
- H_1 ; β_j != 0

Conclusion:-

Based on the output of our model, we conclude that:

- Significant and positive relationships exist between GDPgr and the following variables: BMgr, M0gr, WAGEgr, Unemployment, PPIgr, and EURRUB. For a one-unit increase in BMgr, GDPgr increases by an average of 0.036 (p-value = 0.605). Similarly, for a one-unit increase in M0gr, WAGEgr, Unemployment, PPIgr, and EURRUB, GDPgr increases by 0.02996, 0.027, 0.235, 0.144, and 0.00978 respectively. (p-values = 0.4877, 0.1871, 0.1747, 0.0122, 0.3741).
- Significant and negative relationships exist between GDPgr and the following variables:
  CPI, Interest, and OILgr. For one unit increase in CPI, GDPgr decreases by an average of 0.004075 (p-value = 0.9765). Similarly, for one unit increase in Interest, OILgr, GDPgr decreases by 0.2597, 0.0356 respectively (p-value = 6.77e-05, 6.97e-08).
- Intercept value shows a positive significant relationship of 0.725. (p-value = 0.9576)
- The p-value = 6.941e-09. The null hypothesis is rejected, so we conclude that our model is better than a model with only the intercept because at least one coefficient β is significantly different from 0.
- $\boldsymbol{R^2}$ for this model is 0.6014, which means that 60.14% of the GDPgr of the Russian Economy is explained by independent variables applied in the Model 3 from the given data set.

Model 3 exhibits violations of the assumptions of linearity and homogeneity. To address these issues, we will apply various transformations, such as log transformation, square root transformation, and inverse transformation. By exploring these transformations, we aim to identify the one that provides a better explanation for the GDP growth rate (GDPgr) of the Russian economy.


Posterior Predictive Check
Model-predicted lines should resemble observed data line
Density
GDPgr
Model-predicted data
Observed data
Observed data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (hii)
0.9
0.9
63
46
61
15
62
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
Standard Normal Distribution Quantiles

Multiple Linear Regression Model 4

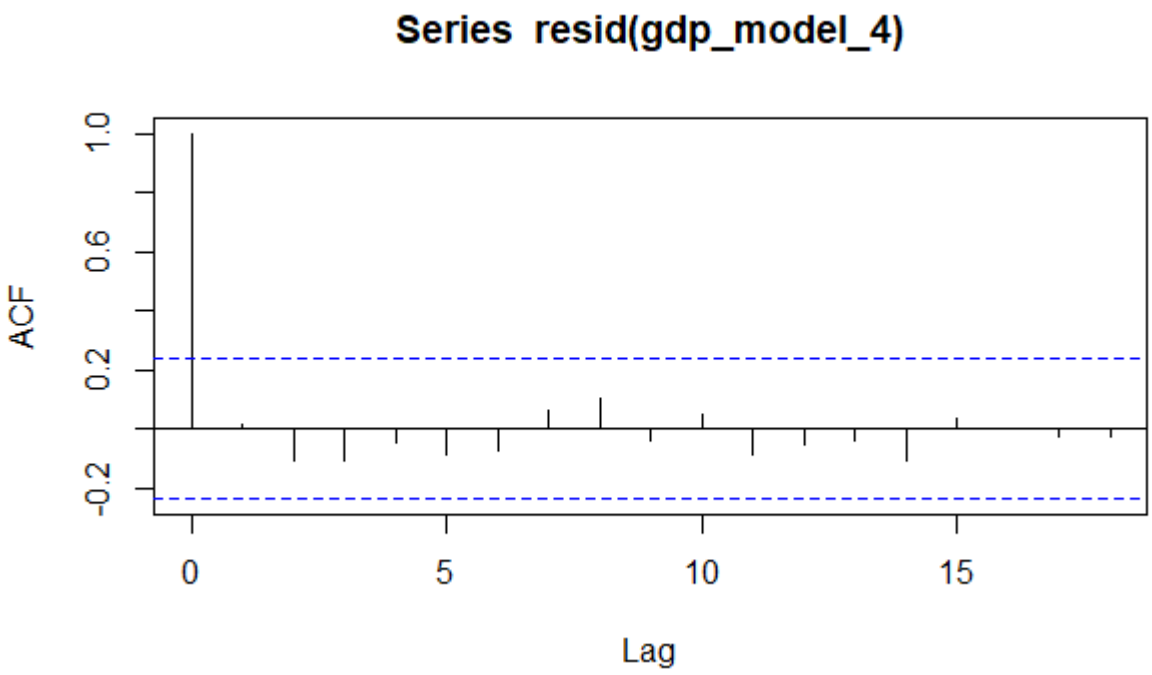
```
#Applying log Transformation on data set for linearity and
Homogeneity of Variances.


constant_value <- abs(min(reduced_data_set)) + 1

trans_data_set <- reduced_data_set + constant_value

log_data_set <- log(trans_data_set)

gdp_model_4 <- lm(GDPgr ~ ., data = log_data_set)

summary(gdp_model_4)
```

```
> summary(gdp_model_4)

Call:
lm(formula = GDPgr ~ ., data = log_data_set)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.117213 -0.006084  0.001812  0.009620  0.033519 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)   
(Intercept)   4.856832   1.991092   2.439  0.01780 * 
BMgr         -0.022279   0.095873  -0.232  0.81706   
CPI          -0.196820   0.409216  -0.481  0.63235   
Interest     -0.128773   0.084110  -1.531  0.13120   
M0gr          0.085498   0.056284   1.519  0.13419   
OILgr        -0.005803   0.004155  -1.397  0.16784   
WAGEgr        0.021774   0.027835   0.782  0.43725   
Unemployment -0.019910   0.239979  -0.083  0.93416   
PPIgr         0.223864   0.074079   3.022  0.00373 **
EURRUB       -0.025742   0.023631  -1.089  0.28051   
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01975 on 58 degrees of freedom
Multiple R-squared:  0.3478,	Adjusted R-squared:  0.2466 
F-statistic: 3.436 on 9 and 58 DF,  p-value: 0.00189
```

Checking the Assumptions/Conditions for the above Model in R programming.

```
#Examining the Assumptions of 4th Multiple Linear Regression
Model

library(performance)

library(lmtest)


residuals_gm_4 <- resid(gdp_model_4)

acf(residuals_gm_4, type = "correlation")

dwtest(gdp_model_4)

check_model(gdp_model_4)
```

```
Durbin-Watson test

data:  gdp_model_t4
DW = 1.9537, p-value = 0.3133
alternative hypothesis: true autocorrelation is greater than 0
```



Based on the diagnostic plots below, we see that:

- Linearity assumption seems valid in the model 4. (Top right plot)
- Homogeneity of variances is still not respected. (Middle left plot)
- Only one outlier exists in the model. (Middle right plot)

- VIF is less than or equal to 5 indicate that Collinearity is not an issue in this model. (Bottom left plot)
- Normality of residuals is respected. (Bottom right plot)
- The p-value associated with the Durbin-Watson test is 0.313, which is above the conventional significance level of 0.05. Therefore, we do not have sufficient evidence to reject the null hypothesis. The null hypothesis states that there is no autocorrelation in the residuals. The alternative hypothesis suggests that there is autocorrelation, specifically a positive correlation between the residuals of adjacent observations. However, since the p-value is greater than 0.05, we do not have enough evidence to support the presence of significant autocorrelation.

Interpretations of coefficients β for Model 4

The table Coefficients gives the estimate for each parameter (column Estimate), together with the p-value of the nullity of the parameter (column Pr(>|t|)).

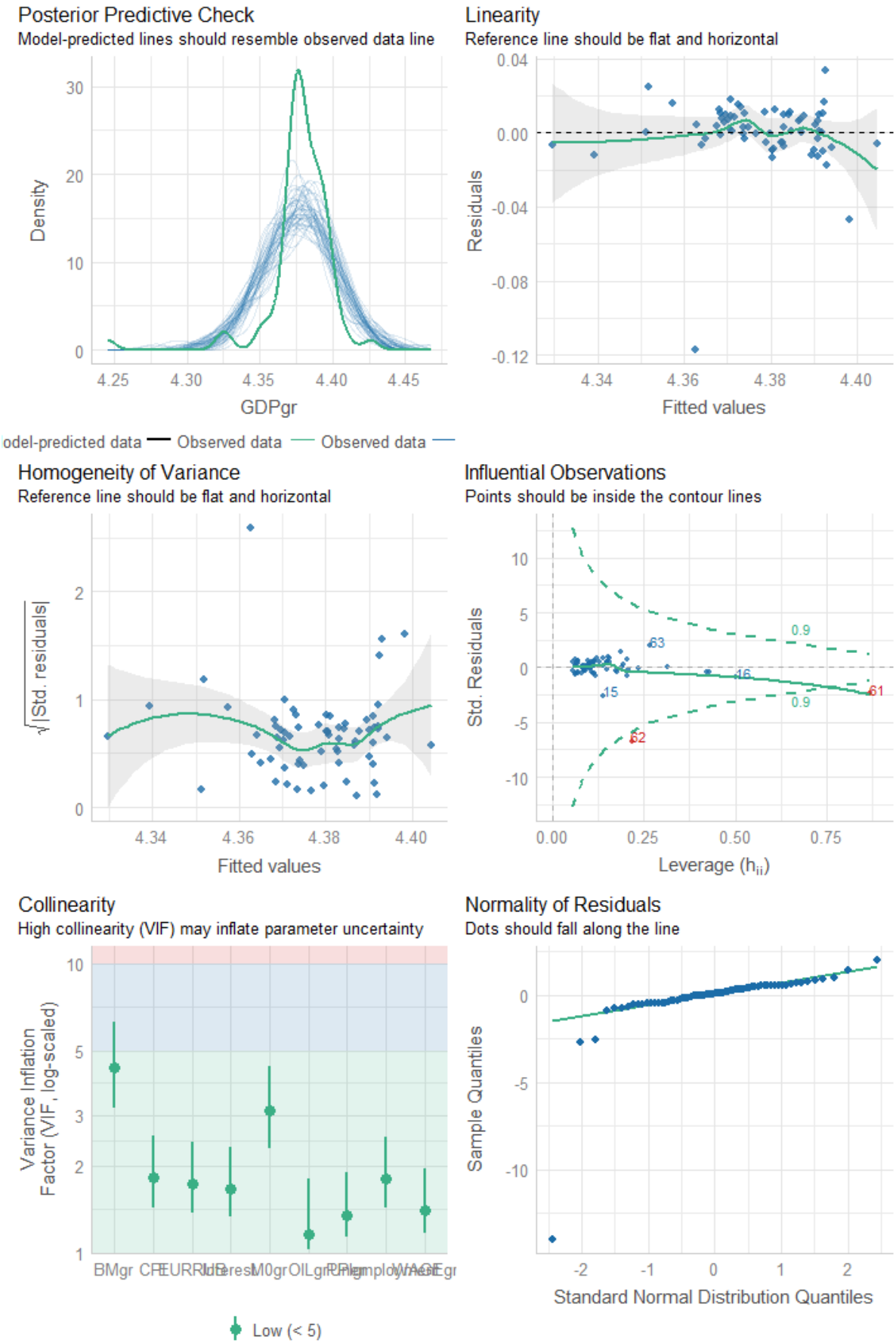
The Test Hypothesis are:-

- H_0 ; $\beta_j = 0$
- H_1 ; β_j != 0

Conclusion:-

Based on the output of our model, we conclude that:

- Significant and positive relationships exist between GDPgr and the following variables: M0gr, PPIgr, and WAGEgr. For a one-unit increase in M0gr, GDPgr increases by an average of 0.0855 (p-value = 0.13142). Similarly, for a one-unit increase in PPIgr, and WAGEgr, and GDPgr increases by 0.224, and 0.021774 respectively. (p-values = 0.00373, 0.43725).
- Significant and negative relationships exist between GDPgr and the following variables:
  BMgr, CPI, Interest, OILgr, Unemployment, and EURRUB. For one unit increase in BMgr, GDPgr decreases by an average of 0.22279 (p-value = 0.817). Similarly, for one unit increase in CPI, Interest, OILgr, Unemployment, and EURRUB decreases by 0.1968, 0.12877, 0.0058, 0.0199, and 0.0257 respectively (p-value = 0.63, 0.131, 0.1678, 0.93416, 0.28).
- Intercept value shows a positive significant relationship of 4.8568. (p-value = 0.0178)
- The p-value = 0.00189. The null hypothesis is rejected, so we conclude that our model is better than a model with only the intercept because at least one coefficient β is significantly different from 0.
- $\boldsymbol{R^2}$ for this model is 0.3478, which means that 34.78% of the GDPgr of the Russian Economy is explained by independent variables applied in the Model 4 from the given data set.


Posterior Predictive Check
Model-predicted lines should resemble observed data line
Density
GDPgr
odel-predicted data
Observed data
Observed data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (h_ii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
Standard Normal Distribution Quantiles

Model 4 from the log transformed data set still exhibits violation of homogeniety of variances. Moreover R^2 is 0.3478 which is critically low to explain the model.

Multiple Linear Regression Model 5

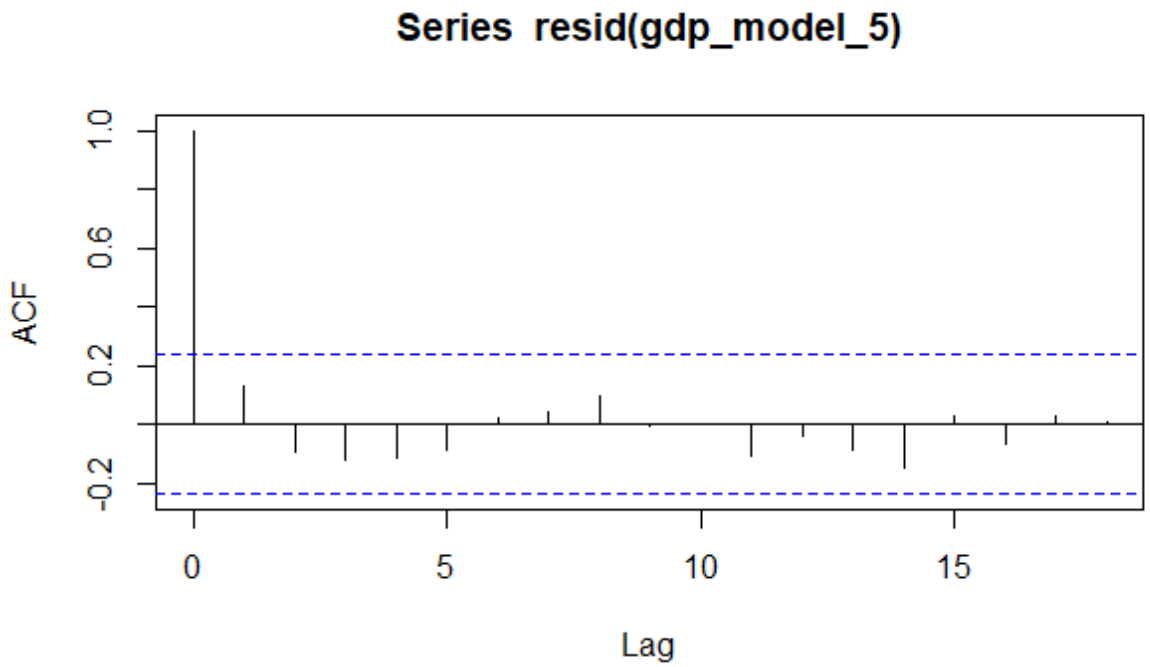
```
#Applying square root Transformation on data set for linearity
and Homogeneity of Variances.

constant_value <- abs(min(reduced_data_set)) + 1

trans_data_set <- reduced_data_set + constant_value

sq_data_set <- sqrt(trans_data_set)


gdp_model_5 <- lm(GDPgr ~ ., data = sq_data_set)

summary(gdp_model_5)
```

```
> summary(gdp_model_5)

Call:
lm(formula = GDPgr ~ ., data = sq_data_set)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.41209 -0.02484  0.01140  0.03516  0.13454 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   8.301747   3.197325   2.596 0.011915 *  
BMgr          0.010390   0.085376   0.122 0.903556    
CPI          -0.073300   0.247146  -0.297 0.767842    
Interest     -0.204929   0.075730  -2.706 0.008927 ** 
MOgr          0.054143   0.051581   1.050 0.298227    
OILgr        -0.022929   0.006605  -3.471 0.000984 ***
WAGEgr        0.024042   0.024834   0.968 0.337005    
Unemployment  0.128412   0.212721   0.604 0.548422    
PPIgr         0.198041   0.066409   2.982 0.004180 ** 
EURRUB       -0.003671   0.016867  -0.218 0.828474    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07976 on 58 degrees of freedom
Multiple R-squared:  0.4473,	Adjusted R-squared:  0.3615 
F-statistic: 5.215 on 9 and 58 DF,  p-value: 3.464e-05
```

Checking the Assumptions/Conditions for the above Model in R programming.

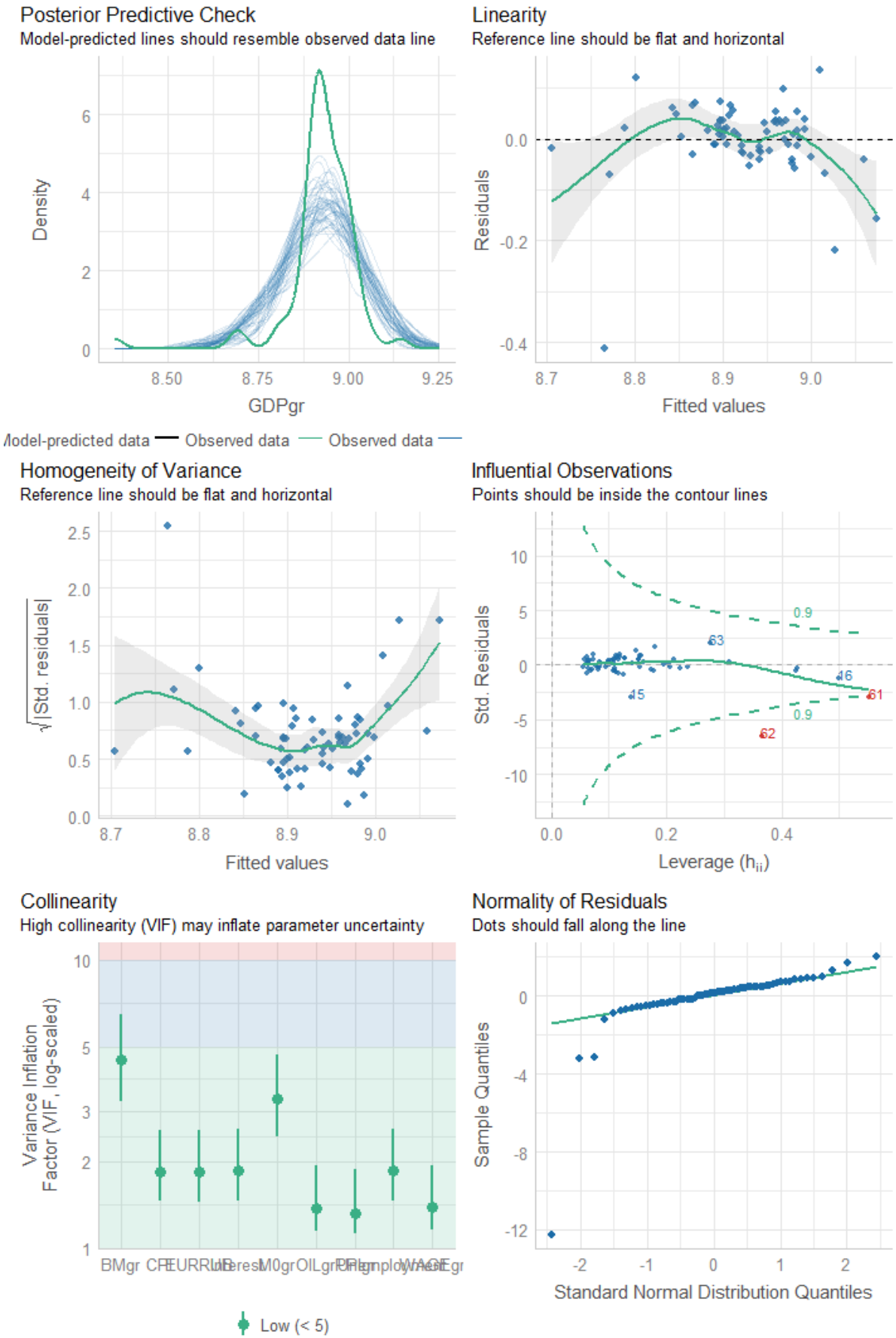
```
#Examine the Assumptions of 5th MLR model

acf(resid(gdp_model_5), type = "correlation")

dwtest(gdp_model_5)

check_model(gdp_model_5)
```

```
Durbin-Watson test

data:  gdp_model_5
DW = 1.7347, p-value = 0.08561
alternative hypothesis: true autocorrelation is greater than 0
```



Based on the diagnostic plots below, we see that:

- Linearity assumption seems not respected in the model 5. (Top right plot)
- Homogeneity of variances is still not respected. (Middle left plot)
- Only one outlier exists in the model. (Middle right plot)
- VIF is less than or equal to 5 indicate that Collinearity is not an issue in this model. (Bottom left plot)
- Normality of residuals is respected. (Bottom right plot)
- The p-value associated with the Durbin-Watson test is 0.856, which is above the conventional significance level of 0.05. Therefore, we do not have sufficient evidence to reject the null hypothesis. The null hypothesis states that there is no autocorrelation in the residuals. The alternative hypothesis suggests that there is autocorrelation, specifically a positive correlation between the residuals of adjacent observations. However, since the p-value is greater than 0.05, we do not have enough evidence to support the presence of significant autocorrelation.


Posterior Predictive Check
Model-predicted lines should resemble observed data line
Density
GDPgr
Model-predicted data
Observed data
Observed data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
Standard Normal Distribution Quantiles

Interpretations of coefficients β for Model 5

The table Coefficients gives the estimate for each parameter (column Estimate), together with the p-value of the nullity of the parameter (column Pr(>|t|)).

The Test Hypothesis are:-

- H_0 ; $\beta_j = 0$
- H_1 ; β_j != 0

Conclusion:-

Based on the output of our model, we conclude that:

- Significant and positive relationships exist between GDPgr and the following variables: BMgr, M0gr, WAGEgr, Unemployment, and PPIgr. For a one-unit increase in BMgr, GDPgr increases by an average of 0.01039 (p-value = 0.90356). Similarly, for a one-unit increase in M0gr, WAGEgr, Unemployment, and PPIgr, GDPgr increases by 0.054, 0.024, 0.128, and 0.198 respectively. (p-values = 0.298, 0.337, 0.548, 0.00418).
- Significant and negative relationships exist between GDPgr and the following variables:
  CPI, Interest, OILgr, and EURRUB. For one unit increase in CPI, GDPgr decreases by an average of 0.0733 (p-value = 0.7678). Similarly, for one unit increase in Interest, OILgr, and EURRUB decreases by 0.205, 0.0229, and 0.00367 respectively (p-value = 0.0089, 0.00098, and 0.8285).
- Intercept value shows a positive significant relationship of 8.3. (p-value = 0.0119)
- The p-value = 3.464e-05. The null hypothesis is rejected, so we conclude that our model is better than a model with only the intercept because at least one coefficient β is significantly different from 0.
- $\boldsymbol{R^2}$ for this model is 0.4473, which means that 44.73% of the GDPgr of the Russian Economy is explained by independent variables applied in the Model 5 from the given data set.

Model 5 from the square root transformed data set gives better result for $\boldsymbol{R^2}$ (0.4473), but seriously violates linearity and homogineity of variances assumption.

Multiple Linear Regression Model 6

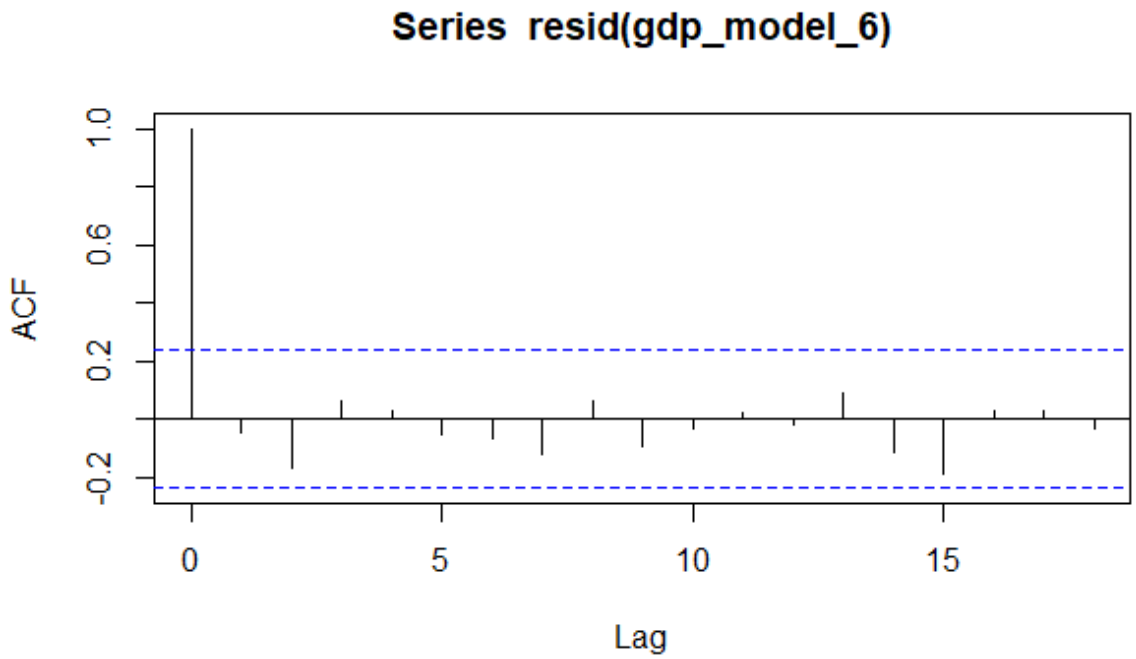
```
#Applying inverse Transformation on data set for linearity and
Homogeneity of Variances.

inverse_data_set <- 1/reduced_data_set

gdp_model_6 <- lm(GDPgr ~ ., data = inverse_data_set)

summary(gdp_model_6)
```

```
> summary(gdp_model_6)

Call:
lm(formula = GDPgr ~ ., data = inverse_data_set)

Residuals:
     Min       1Q   Median       3Q      Max 
-142.701   -1.764    2.334    5.020   33.578 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)  
(Intercept)  -1.287e+02  1.988e+02  -0.647   0.5201  
BMgr         -9.514e-01  1.956e+00  -0.486   0.6285  
CPI           1.498e+04  2.070e+04   0.724   0.4722  
Interest      2.514e+00  6.082e+01   0.041   0.9672  
MOgr         -3.629e-01  1.046e+00  -0.347   0.7299  
OILgr         2.745e-02  3.329e+00   0.008   0.9934  
WAGEgr        1.290e+00  8.570e+00   0.151   0.8808  
Unemployment -5.937e+01  1.207e+02  -0.492   0.6245  
PPIgr        -5.772e+00  2.380e+00  -2.425   0.0185 *
EURRUB       -3.092e+02  5.451e+02  -0.567   0.5727  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.93 on 58 degrees of freedom
Multiple R-squared:  0.1132,	Adjusted R-squared:  -0.02438 
F-statistic: 0.8228 on 9 and 58 DF,  p-value: 0.5977
```

Checking the Assumptions/Conditions for the above Model in R programming.

```
#Examine the Assumptions of 5th MLR model

acf(resid(gdp_model_6), type = "correlation")

dwtest(gdp_model_6)

check_model(gdp_model_6)
```

```
Durbin-Watson test

data:  gdp_model_6
DW = 2.0931, p-value = 0.487
alternative hypothesis: true autocorrelation is greater than 0
```



Based on the diagnostic plots below, we see that:

- Linearity assumption seems not respected in the model 5. (Top right plot)
- Homogeneity of variances is still not respected. (Middle left plot)
- Only one outlier exists in the model. (Middle right plot)
- VIF is less than or equal to 5 indicate that Collinearity is not an issue in this model. (Bottom left plot)
- Normality of residuals is respected. (Bottom right plot)
- The p-value associated with the Durbin-Watson test is 0.487, which is above the conventional significance level of 0.05. Therefore, we do not have sufficient evidence to reject the null hypothesis. The null hypothesis states that there is no autocorrelation in the residuals. The alternative hypothesis suggests that there is autocorrelation, specifically a negative correlation between the residuals of adjacent observations. However, since the p-value is greater than 0.05, we do not have enough evidence to support the presence of significant autocorrelation.

Interpretations of coefficients β for Model 6

The table Coefficients gives the estimate for each parameter (column Estimate), together with the p-value of the nullity of the parameter (column Pr(>|t|)).

The Test Hypothesis are:-

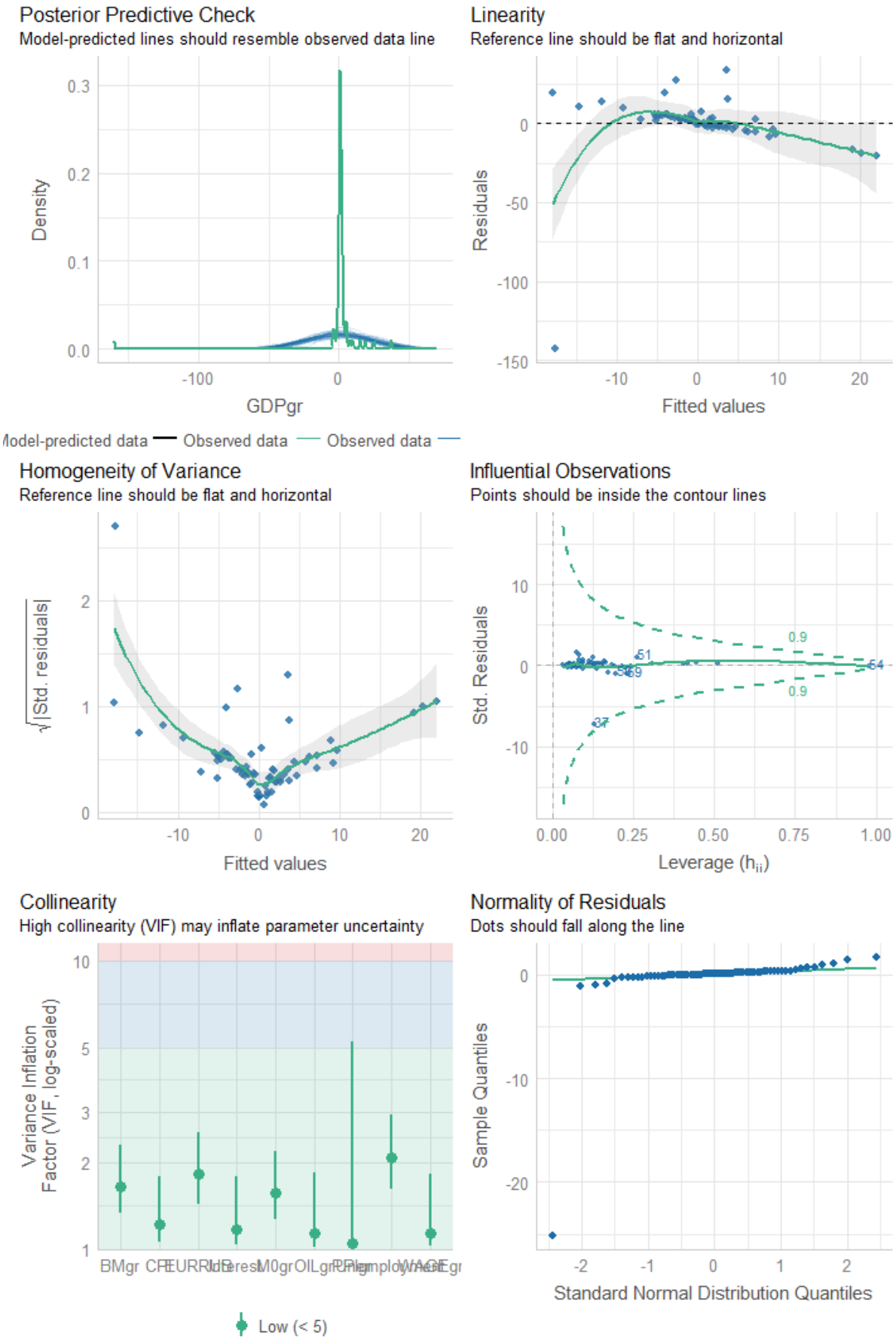
- H_0 ; $\beta_j = 0$
- H_1 ; β_j != 0

Conclusion:-

Based on the output of our model, we conclude that:

- Significant and positive relationships exist between GDPgr and the following variables: CPI, Interest, OILgr, and WAGEgr. For a one-unit increase in CPI, GDPgr increases by an average of 1.498E+04 (p-value = 0.4722). Similarly, for a one-unit increase in Interest, OILgr, and WAGEgr, GDPgr increases by 2.514, 2.745e-02, and 1.29 respectively. (p-values = 0.9672, 0.9934, 0.8808).
- Significant and negative relationships exist between GDPgr and the following variables:
  BMgr, M0gr, Unemployment, PPIgr, and EURRUB. For one unit increase in CPI, GDPgr decreases by an average of 9.514e-01 (p-value = 0.6285). Similarly, for one unit increase in M0gr, Unemployment, PPIgr, and EURRUB decreases by 3.63e-01, 5.937e+01, 5.772, 3.09E+02 respectively (p-value = 0.7299, 0.6245, 0.0185, 0.5727).
- Intercept value shows a negative significant relationship of 1.287e+02. (p-value = 0.52)
- The p-value = 0.5977. The null hypothesis is accepted, so we conclude that model with only the intercept is better, because coefficients β are not significantly different from 0.
- $\boldsymbol{R^2}$ for this model is 0.1132, which means that 11.32% of the GDPgr of the Russian Economy is explained by independent variables applied in the Model 6 from the given data set.

Model 6 from inverse transformed data set gives p-value 0.5977, accepting the null hypothesis β_i = 0, which means that model is better explained by intercept value alone.


Posterior Predictive Check
Model-predicted lines should resemble observed data line
Density
GDPgr
Model-predicted data
Observed data
Observed data
Linearity
Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log-scaled)
Low (< 5)
Normality of Residuals
Dots should fall along the line
Sample Quantiles
Standard Normal Distribution Quantiles

Conclusion Remarks

In this document, I conducted comprehensive testing of Multiple Regression Modelling on the Russian Econmic Data Set, using R Studio RStudio 2023.03.1+446. I applied 6 different Regression Models to identify better model that explains GDPgr (Gross Domestic Product growth rate), which refers to the rate at which the Gross Domestic Product (GDP) of a country or region is changing over a specific period of time. It measures the percentage increase or decrease in the value of goods and services produced within the economy compared to the previous period. The GDP growth rate is a key indicator of economic performance and reflects the overall health and expansion of an economy.

Data Set consisted of 20 variables, with 68 total observations. Time and Date variables were not required for MLR model. After dropping those two variables, MLR models were applied to remaining 18 variables.

Summary of the Models are given below;

- The very 1st MLR model was applied with all 17 the factor variables included to determine GDPgr variable. Model explained 72% of the GDPgr with a p-value of 2.49e-09, violating almost all the assumptions of MLR.
- The 2nd MLR model was applied as step wise regression model technique. It explained almost 72% of the GDPgr with a p-value of 1.02e-11. Assumptions of Normality and Autocorrelation was respected.
- The 3rd MLR model was applied with theoretical knowledge and dependence of GDPgr on the given variables in the data set. It explained 60 % of the GDPgr with p-value of 6.94e-09, not respecting the assumptions of linearity and homoginiety of variances.
- The 4th MLR model was applied to minimize the violaiton of linearity and homoginiety of variances assumptions, for that purpose log transformation was applied to data set and then MLR was fitted which gave 34% explaination of the GDPgr. (p-value = 0.00189). All Assumptions were respected but the Homoginiety of variances.
- The 5th MLR model was employed for the aforementioned purpose, using square root transformation, that gave $\boldsymbol{R^2}$ 0.4473 with a p-value of 3.464e-05. But the assumptions of linearity and homogienty were not met.
- Finally fitted the 6th model, using inverse transformation. Which gave p-value of 0.5977, accepting the null hypothesis of $\beta_i = 0$. This inverse transformed data set model was better explained by only the intercept value.

None of the above applied model better explaines the GDPgr. It might be necessary to explore other regression models that are more suitable for this data. This could include nonlinear regression, generalized linear models, or machine learning algorithms like decision trees or support vector machines. Scope of these models are out of the given assignment.

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